



DEEDS BARN
and the
SELF-STARTER



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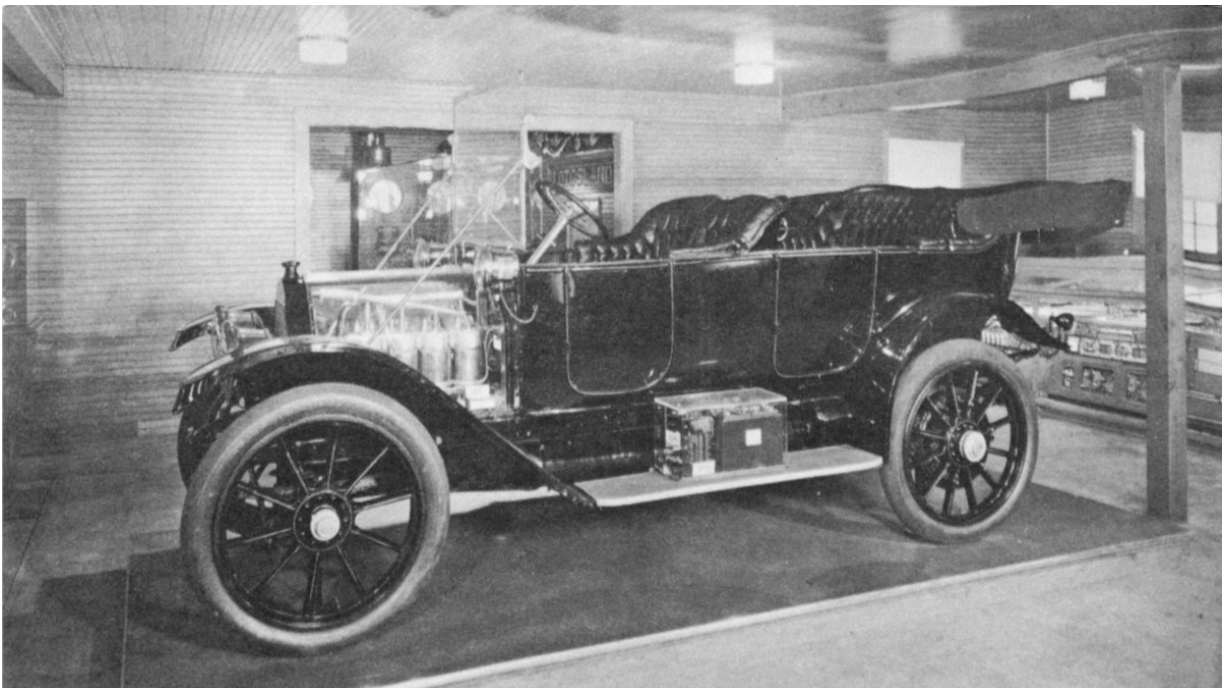
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*** START OF THE PROJECT GUTENBERG EBOOK DEEDS BARN
AND THE SELF STARTER ***

DEEDS BARN

and the

SELF-STARTER



The 1912 Cadillac, equipped with self-starter, is the central exhibit in Deeds Barn.

Deeds Barn is symbolic of one of the most important developments in the automotive field. It was here that the modern ignition system for automobiles and the self-starter were born and the foundation laid for one of Dayton's most important industries. The barn itself and one of the 1912 Cadillac cars, first to be commercially equipped with a self-starter, represent this important invention. Also included in the barn are several

other exhibits which should be of interest to any visitor and most certainly to every Daytonian.



The original Deeds Barn was erected at 319 Central Avenue, Dayton. An exact replica stands in Carillon Park.

The Story of the Self-Starter



C. F. KETTERING



E. A. DEEDS

These photos were taken about the time the Delco firm was incorporated.

In industrial production, as in so many other activities, large oaks from little acorns grow. The cash register and the electric self-starter are instances in

point. The first factory of The National Cash Register Company under the presidency of John H. Patterson, was a room in the Callahan Power Building, forty feet wide and eighty feet long, with thirteen people on the payroll. From it emerged a business on which the sun never sets.

More humble in environment was the invention of the self-starter which had its birth in a barn. Almost before a decade passed, it had made the use of the automobile universal. Such were the beginnings of two enterprises which, to a degree greater than any others, enhanced the industrial prestige and prosperity of Dayton. Each contributed a chapter to the romance of American Industry.

Dayton people, and for that matter, a considerable portion of the outside world, are familiar with the story of John H. Patterson's industrial genius that made his name synonymous with cash register production and distribution. They are not so well acquainted with the origin of the self-starter and particularly the historic barn in which it was created. A replica of that barn stands today in Carillon Park.

The Drama of the Barn, as it may well be called, evolved about two men—Edward A. Deeds and Charles F. Kettering. Both were born and reared on the farm, that rugged nursery in which so many of our national leaders were nurtured; both worked their way through college; both began their careers near the bottom of the ladder at The National Cash Register Company. Their business alma mater served them well. 2



The hayloft of the Deeds barn was equipped as a workshop.

Fate must have decreed that Deeds and Kettering should meet. In 1904 Deeds was Superintendent at the NCR. Electrically minded since his boyhood on the farm, he had an unwavering confidence in the application of electricity to industry. At that time the cash register was operated by turning a crank. Deeds conceived the idea of putting a motor on the machines to supersede the crank. He designed and built the first motor to be employed for this use. This is an important link in the Deeds-Kettering story, because it brought the two men together.

Deeds was absorbed in many responsibilities at the NCR and could not devote the necessary time to the perfection of the motor. He wanted a bright young electrical engineer to whom the electrification project could be entrusted. He asked Alfred D. Cole, his old professor of physics at Denison University, to suggest a likely candidate. Cole, who had joined the faculty of Ohio State University, suggested Kettering, his star student, who was about to graduate. Deeds wrote to Kettering, offering him a position at \$50 a week and the offer was accepted, not, however, without some qualms on Kettering's part.

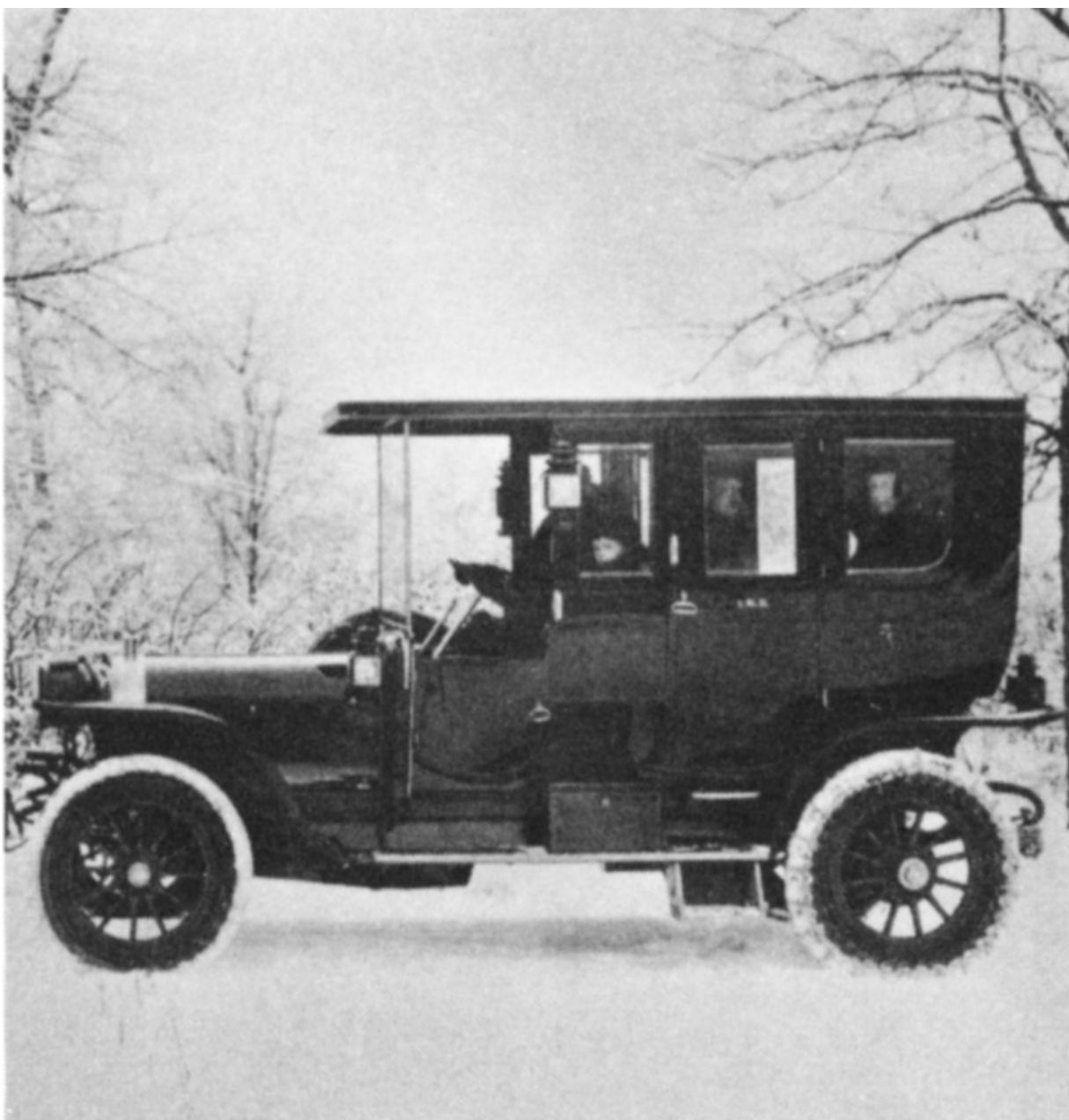
When Kettering took over his job in the Inventions Division at the NCR it was to mean more than a brilliant contribution to the development of the National cash register. It marked the beginning of a kinship with Deeds that was to make inventive history. The reason for this kinship did not rest 3 solely on the fact that they were both electrically minded. Also important was the fact that they embodied certain opposites of talent and temperament which so often make for success.

In 1904, when Deeds and Kettering met, the gasoline-propelled vehicle, despite jests, jokes and jeers, was becoming a practical reality. Great inventors like Thomas A. Edison and Charles P. Steinmetz, however, insisted that it was doomed to failure, contending that electricity was the one and only effective motive force for the horseless carriage. Deeds believed otherwise. He saw the internal combustion engine as the driving power of the future. He had been so impressed with the gasoline automobile that for a time he had an ambition to go into its production. Unable to do this, he decided to “put something”, as he expressed it, on motor cars. This ambition pointed the way to what became the electric ignition, lighting, and starting system.

In order to put that “something” on a car it was necessary to have a car. Here is where the famous barn first entered the picture. Deeds lived at 319 Central Avenue in Dayton. Back of the house was a barn which housed a horse and phaeton. Deeds could easily have purchased an automobile had he wished. A purchased car, however, would not serve the purpose that clamored for action. That “something” could only be placed on an automobile that he literally knew from the ground up. With the aid of Fred Schmitt, an expert machinist, he built and assembled a car which was named “Suburban Sixty.” All the work on it was done in the barn. Deeds put a two-speed axle on the barn product but found that it did not function satisfactorily. He therefore substituted an auxiliary transmission which did work.



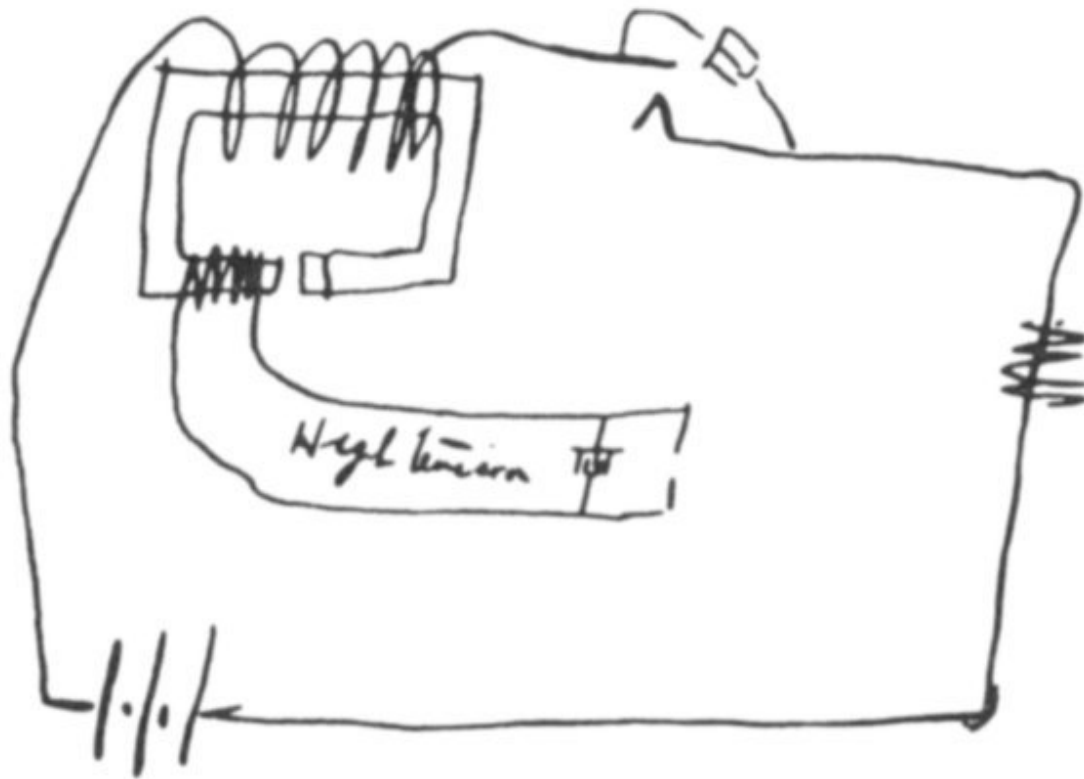
The "Suburban Sixty," built by E. A. Deeds, with the summer and winter bodies which made it an all-weather model.



Deeds now had a car and the germ of an idea. Once more, absorption in responsibilities at the NCR prevented him from devoting sufficient time to the development of the idea. As he pondered over what to do, he recalled an experience that he had while building the Shredded Wheat plant at 4 Niagara Falls. In order to obtain some equipment for a miniature electric railway which was to be installed in the factory restaurant, he was obliged to deal with two men who operated a small manufacturing firm in Providence. These men sat in an unpretentious office with a desk between

them and operated on a fifty-fifty basis. One ran the factory; the other looked after business details.

July 23 d 08
Ignition System.

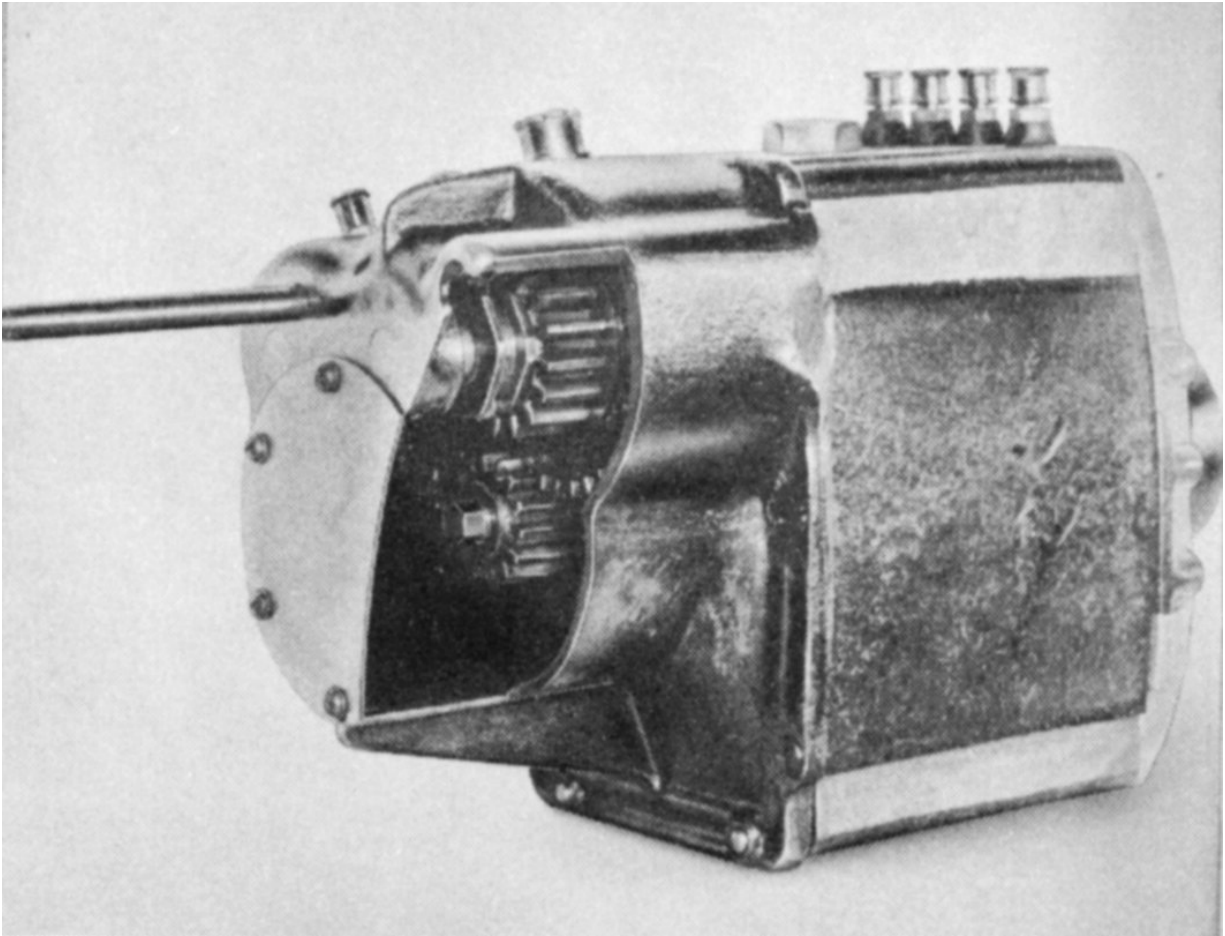


C. F. Kettering

...

Witnesses
B. A. ...
P. M. ...

This crude drawing was the basis of the original patent issued to C. F. Kettering.



A model of the self-starter as it was built in 1912 and furnished to Cadillac.

Deeds and Kettering had often talked of the great things they were going to do. On one occasion when they sat on a red couch in the living room of the Deeds residence, Deeds spoke of the two small Providence manufacturers and their simple plan of ownership and operation. The inspiration came to him: “Why can’t we do the same thing?”

That “something” on a motor car still clamored for development. Vaguely, Deeds saw it as an electric ignition system that would emancipate motorists from one of the tribulations of the early automobile era. The “Suburban Sixty” was equipped with an Atwater-Kent single spark battery ignition. It was efficient but complicated and expensive. Deeds envisaged improved ignition as another electrical possibility.

Kettering, whose very life was keyed to electricity, had been thinking along the same line. Deeds had tremendous confidence in Kettering, born of appreciation of his great gifts which had already registered a solid achievement at the NCR. He had perfected the motor for the cash register and had worked miracles on various machines. In Kettering, Deeds found a heaven-sent associate, animated by the same ideal of purpose as his own and one who could share his dreams and aspirations. When Deeds told Kettering of his ideas, he found a willing listener. On the suggestion that they combine their efforts, the lean, angular inventor needed no urging. 5 He put out his hand, stained by chemicals, and answered:

“That suits me. We’ll shake on it.”

There and then the Deeds-Kettering partnership came into being, unsigned but sealed by mutual faith and cemented by loyal friendship. The informal alliance was fated for an accomplishment that not only opened up new inventive horizons but contributed to the standardizing of automobile production and operation.

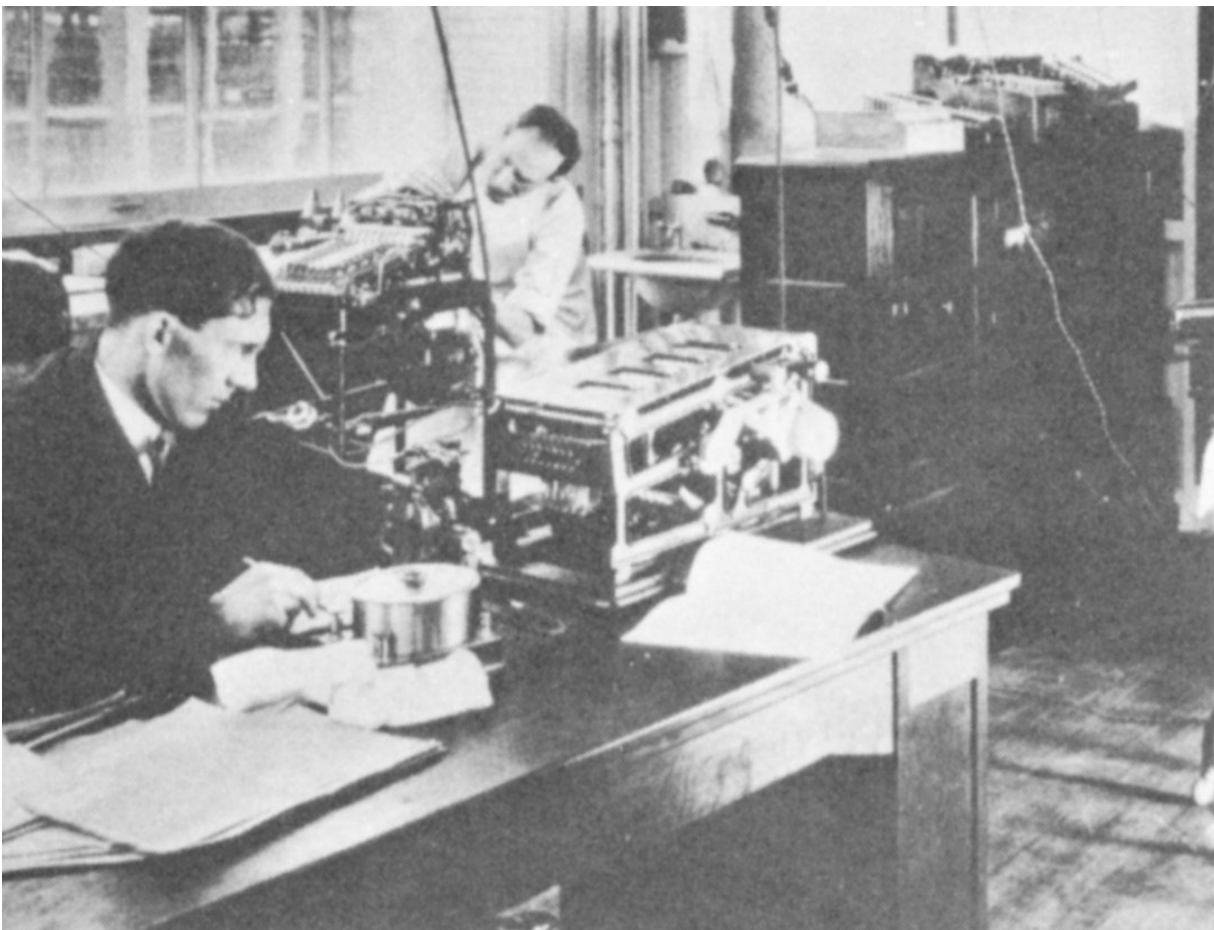
The first objective was a workshop. “Suburban Sixty” had been built in the Deeds barn. Deeds now turned the barn over to the project. The barn became a laboratory in which ideas flashed and sometimes clashed back and forth in almost dazzling succession.

Before the curtain rises on the Drama of the Barn let us envisage the principal actors as they stood on the threshold of the great adventure. No two men could have presented a greater physical contrast. Deeds was ruddy, keen-eyed, broad of shoulder; Kettering was tall, spare, and spectacled. Deeds was the engineer by instinct and training with a flair for finance and organization. Behind him already lay considerable experience as an industrial executive. Thus he became the wise counsellor and negotiator. Kettering was the technical expert with a genius for invention. To scientific imagination he brought a rare insight into mechanism. Deeds had once said, in discussing the development of the electric ignition, lighting, and starting system, “Men must hunt in pairs.” For the enterprise now to be initiated, the ideal “pair” existed. The two men represented an unbeatable combination.



Some of the members of the “Barn Gang,” left to right: W. A. Chryst, W. P. Anderson, C. F. Kettering, E. A. Deeds.

Kettering started work in the barn in the summer of 1908. The equipment was already installed because Deeds had assembled it for the production of “Suburban Sixty.” It consisted of a milling machine, a drill press, an engine lathe, a tool grinder, a forge, one ten-horsepower motor, an oscillograph, a long bench, and small tools. All the machinery was concentrated in the hayloft while the “Suburban Sixty,” the first guinea pig, rested on the lower floor. Because the equipment was in the old hayloft, Kettering was dubbed the “hayloft inventor.” Connecting the two floors was a narrow, winding stairway on which Kettering dashed up and down a hundred times a day. 6



Kettering in the early days when he was in the Inventions Division at NCR.



Kettering after he started to work on the self-starter.

In the early period of the project Kettering and Deeds could work only on weekends, nights, and holidays because they were both employed at the NCR. In 1909 Kettering realized that if the undertaking was to get anywhere, it would be necessary for him to devote all his time to it. He therefore resigned as Chief Inventor and set up shop in the barn, which was practically his domicile for the next two years.

During Kettering's years at the NCR factory, his chief assistant had been William A. Chryst, who had risen from office boy to a position of high skill and responsibility. He became an enthusiastic ally of the project. Soon he made a third to the duet that plugged away in the barn, working as a volunteer.

During the period of incubation of ignition, the first link in the electrification system for the automobile, Deeds largely financed the operation. Kettering's wants were simple. Deeds and Chryst had their posts at the NCR. Nevertheless it was hard financial sledding for everybody concerned. There were many times when the outlook was bleak and the barn treasury low.

Chryst was the first recruit of the group of men who came to be known as the "Barn Gang." Next came William Anderson. As months passed they were joined by Zerbe Bradford, William Mooney, John Reece, Ralph Todd, Harvey I. Phillips, John Lipes, Albert Koffer, John Sheats, Robert Demaree, and W. G. Johns. These men formed the nucleus of what later became the Delco organization.

Back in Denison, Professor Cole had often said to Deeds: "The storage battery will come into its own." He transmitted his faith to Deeds who 7 found an enthusiastic supporter in Kettering. An electric ignition system stirred their imagination. It was to be part of that "something" that Deeds longed to put on a car.

The need of a new and effective ignition system was imperative. For years motorists had struggled with the magneto apparatus which was a prize producer of headaches and backaches. On a magneto it was not possible to run slow in high gear in heavy city traffic without risk of stalling which, when it did occur, meant hopping out of the car and cranking up. To crank a car on a magneto when the dry batteries on the auxiliary system had run down, demanded a considerable amount of endurance. It prevented the women motorists from driving alone.

The system that created many of these complications was equipped with a magneto and an auxiliary vibrating coil dry battery. The basic trouble was that it produced an insufficient amount of current. Hence the frequent stalling. What Kettering did in that long ordeal in the barn was to change from a vibrating system to a single spark which conserved the battery and produced ample electric power for ignition. The vibrators, which demanded endless tinkering every time a car was taken out for use, vanished from the spark coils. Kettering's dictum was: "No spark outside the cylinder but a good one inside." To prove the efficiency of the electric ignition system,

Deeds, accompanied by Mrs. Deeds, drove to New York and back on one set of dry batteries, which were still not exhausted when the car arrived back in Dayton.

With the electric ignition complete, the big question was to whom it should be submitted. It was not a difficult problem once the motor manufacturing field was surveyed. At the head of the list of producers was Henry M. Leland, whose name had become synonymous with Cadillac. “Uncle Henry,” as Leland was affectionately known, was the Grand Old Man of the automobile industry. It was a tradition in the business that “if you can satisfy ‘Uncle Henry’ you can meet the highest and most exacting requirements.” He was an engineer of international reputation and the Cadillac was the apple of his eye.

As was the case with every other automobile manufacturer, Leland was fed up with the magneto ignition. When the barn product was brought to his attention he was interested. Chief Engineer Sweet of the Cadillac Company came to Dayton to investigate. Deeds and Kettering met him at the railway station and took him for a ride in Kettering’s Cadillac roadster which was equipped with the new electric ignition. They covered miles of Dayton streets and more miles of rough country roads adjacent. Then they landed Sweet safely at the station. When they tried to start up the car after his departure, the system stalled. A wire had broken in cooling off and fell apart. If the system had broken down while Sweet was aboard the car, 8 events might have turned out differently.



A scene duplicated many times—W. A. Chryst and C. F. Kettering “trying it out” on an actual run.

When an ignition set was sent to Detroit, Leland shared Sweet’s favorable impression. Negotiations for a contract now began by correspondence. The great day came when Mr. and Mrs. Deeds and Kettering went to Detroit for the signing of the contract. Leland was immensely taken with the two eager, earnest young men who walked into his office that autumn day in 1909. The deal was threshed out in a discussion that lasted for some hours. Then a contract was signed for 5,000 ignition sets.

Now developed something of a dilemma. Deeds and Kettering had no firm name, in fact no formal business association at all. The contract therefore was made out in Deeds’ name. Deeds and Kettering started off from the Cadillac plant in a borrowed car. Soon after they got under way it was discovered that the precious contract was in Kettering’s pocket. Taking it out he said, “I don’t want the danged thing. You keep it.” With this he handed it over to Deeds. Soon he left the car and Deeds proceeded to the home of an old friend where Mrs. Deeds was anxiously awaiting the outcome of the negotiations. In his full and fruitful life Deeds was to know many precious hours. But no moment was sweeter than when he handed the contract to his devoted companion of the years with the words, “It’s signed.”

The Cadillac contract presented another problem for Deeds and Kettering because they had no plant. Their original idea in uniting for the barn 9 project was to develop what Kettering called an “idea hatchery.” Production had never been considered. The problem was soon solved. While working during vacations for the Star Telephone Company of Ashland, Ohio, Kettering had had some dealings with the Kellogg Switch-Board and Supply Company of Chicago. He got in touch with them and they agreed to manufacture the sets.

Deeds and Kettering now had a contract, a commercial product, and a firm to make it. The next step was to organize a corporation with a striking name. Deeds suggested The Dayton Laboratories and Engineering Company. When Kettering asked Chryst what he thought of it, he got this reply:

“Not so good. Why don’t you change the words around and call it The Dayton Engineering Laboratories Company?” This time Kettering was unimpressed. But Chryst’s suggestion prevailed and the company was incorporated July 22, 1909. Remembering the experience of the two small Providence manufacturers, Deeds projected the company on a fifty-fifty basis with a capital stock of \$150,000, two-thirds of which was common stock. The shares were equally divided between the two men. Each was a joint owner with the other in all contracts and patents. Deeds became President and Kettering Vice President. With incorporation, a modest office was opened in the United Brethren Building on Main Street with a staff consisting of one man who was clerk, bookkeeper, secretary, and general factotum.

Chryst had succeeded Kettering in the NCR Inventions Division, but spent every moment of his spare time at the barn. He was immensely proud of his choice of the company name. As he worked about the barn he began to refer to it as “Delco.” Soon Deeds, Kettering and the rest of the “Barn Gang” took it up. In this way the name that became a household word came into being.

Then trouble broke. Soon after the first ignition sets had been delivered to Cadillac, Kettering received a telegram from Leland saying that the controlling relays, the heart of the ignition system, were sticking. “Come at

once” was Leland’s command. Kettering was certain that the specifications given to the Kellogg Company were accurate. He went to Detroit, tried out the system, and sure enough the controlling relays stuck. Leland had announced that the new ignition system would be placed on his 1910 car. He prided himself on the integrity of his word. It was up to the “hayloft wizard” to make good.

Kettering put the relays in his bag and took the midnight train back to Dayton. The next twenty-four hours would be crucial for Delco. As sleep evaded him he reached into his bag, got out one of the relays, and began to feel the pole pieces and the armature. Then, in the darkness of his 10 upper berth, he discovered what was wrong. The Kellogg people had made the pole pieces a trifle rounded instead of perfectly flat. Kettering went straight to the barn from the railway station, machined the pole pieces down, and returned to Detroit on the next train with one source of trouble ended. Another anxiety developed when there was coil trouble. Again Kettering dashed to Detroit and worked a miracle.

The spark ignition, with the slogan, “Nothing on the dashboard but the switch,” went on the 1910 Cadillac and operated to the satisfaction of every user. A new cycle of the Motor Age dawned. Delco was launched.

A trouble far more serious than mechanical anxieties now developed. In his almost vicious assault on the barn problems, Kettering strained his eyesight. It augmented a weakness that had started in his college days when he burned the midnight oil. For a time it was feared that his vision would be permanently impaired. Despite this handicap he stuck resolutely to the job.

When the ignition contract was signed with Cadillac there were 308,930 automobiles and 6,030 trucks in the United States. These totals represented a tremendous advance from the day when it was a debated question as to whether the horseless carriage would be propelled by steam, electricity, or gasoline. Gasoline had won out and the internal combustion engine came into its own.



Another view of the hayloft workshop, W. A. Chryst with back to camera.

Although the automobile had “arrived” it still labored under a serious handicap. It was necessary to crank a car by hand in order to start it. Often it was a back-breaking and sometimes an arm-breaking operation. The 11 crank still made the motor car something of a butt for cartoonists and jokesters.

Once more, Leland stepped into the Delco picture. Like every other automobile manufacturer, he had a grievance against the crank. One day he picked up his friend, Carter, while driving to the Cadillac plant. On the Belle Isle bridge the car stalled. Carter got out to crank it; the engine kicked back and broke his arm.

Leland sent for Kettering and told him of the Carter mishap. Then he said:

“The crank is vicious, turbulent and unruly. I am breaking arms all over the country and it’s got to stop. In the past six months six of my workmen have fractured their arms while cranking cars. What can be done?”

Kettering replied:

“I believe that it would not be difficult to crank a car by electricity.”

Always alert for improvement and never hesitating to do pioneering, Leland said:

“If you can provide an electric starter, I will put it on my 1912 car.”

There was nothing particularly new about the self-starter as such. There were starters that operated by a coiled spring, and starters that used gas or compressed air. None of them, however, had come into anything like wide and satisfactory use. Electricity had turned the trick for ignition; Kettering was certain it could repeat the performance for a self-starter.

Kettering returned to Dayton and on the same train with him went a new Cadillac engine. Both were in the barn before day dawned. Deeds appeared to lend a hand. They roused out an implement maker, got him to send over sprocket wheels and chains, and rigged a temporary set-up with motor and storage batteries. Now came the great question, “Can we do it electrically? Will it turn the engine over?” The “Barn Gang” worked all day, Deeds calm and imperturbable; Kettering with his customary dynamic energy. Late in the afternoon Kettering said, “I wonder if the danged thing will work,” to which Deeds replied, “We’ll soon find out.”

The connection was made and the engine turned over. “Then,” as Deeds commented later on, “we knew that quantitatively it was practical.” It was a memorable hour, for in the gathering dusk was born the second of the barn products that put the cap-stone on motoring, and made it safe as well as useful. With the perfection of the electric self-starter, a woman, even a child, could get into a car knowing that by merely pressing a button, the machine would start. It meant farewell to arm- and back-breaking. Gone was the crank and gone, too, was the acetylene tank of compressed gas for the headlights. The electric lighting of automobiles came simultaneously with the self-starter. 12

Convinced from his tests of sample sets that the electric self-starter was practical, Leland announced that the new device would go on his 1912 car. It was now late in 1910. Time again was of the essence for work in the barn, and time was beginning to run out. As with ignition, trouble came. When

the new Cadillac car arrived at the barn it was discovered that the engineers at Detroit had not left sufficient room for the self-starter. This meant new drawings and new designs. With this obstacle hurdled, disaster broke. While driving on a stretch of bad road between Dayton and Xenia, Kettering's car slid into a ditch. The impact not only smashed the car but broke one of his ankles. The foot was put into a cast and Kettering was ordered to remain in bed for two weeks.

On the morning after the accident a telegram from Detroit brought the news that the only Cadillac car equipped with one of the electric self-starters had caught fire in Leland's garage. The starter was badly damaged. Within twenty-four hours Kettering was under that car, his crutches on the ground alongside, and his cast-bound leg sticking out from beneath the chassis. He brought the bruised and blackened starter back to life and action.

Now developed a new source of anxiety. Battery manufacturers, cautious and skeptical, maintained that no electric storage battery could be produced that would meet the requirements imposed. Delco needed thousands of batteries. In their dilemma a new ally came to their aid. He was O. Lee Harrison, salesman for the Electric Storage and Battery Company of Philadelphia, one of the largest concerns of its kind in the country.

He worked out of the Cleveland branch and while at Steubenville, Ohio he received this wire from the manager:

"We have a letter from the Dayton Engineering Laboratories Company signed by a man named Kettering who wants to buy 5,000 storage batteries. Go down and see what he looks like."

Harrison went to the little Delco office in the United Brethren Building. Wearing an old golf cap, Kettering was seated in a swivel chair with his feet propped up on the top of a desk. To him Harrison said:

"I am looking for a man named Kettering who wants to buy 5,000 storage batteries. I don't expect to sell him but I want to see what he looks like."

The man at the desk dropped his feet to the floor and said:

“I am Kettering.”

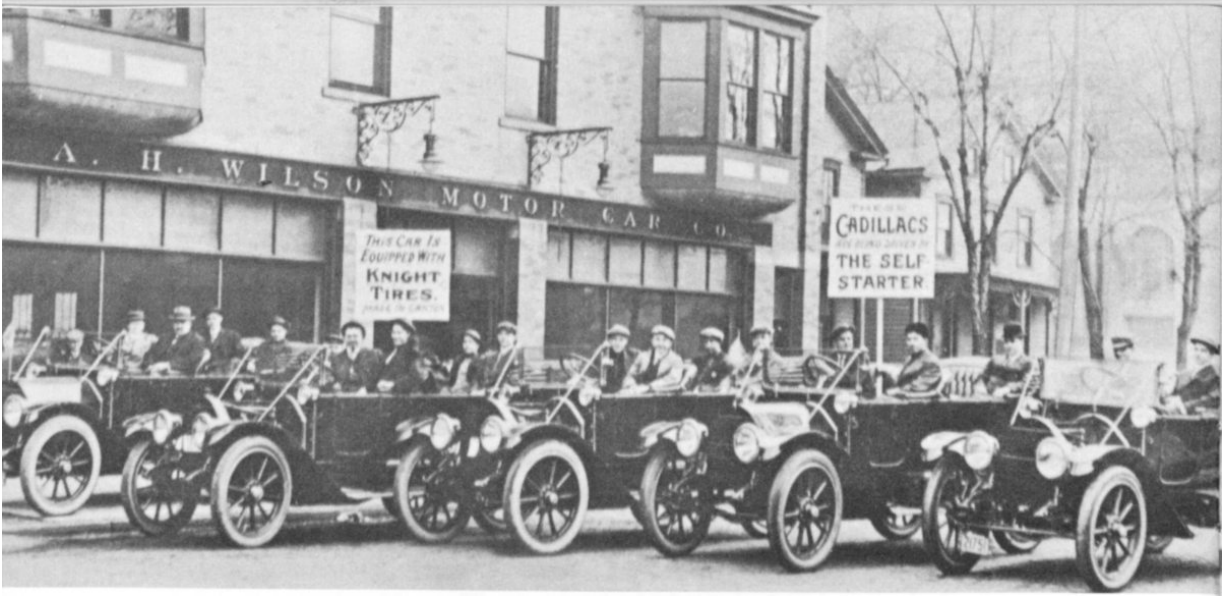
Kettering took Harrison to the barn and showed him the self-starter. In its original form it had sprockets and wheels and looked like a 13 McCormick binder. When Harrison said it reminded him of the farm, Kettering remarked that he, too, was a farm boy. From that moment the two men clicked. Harrison went to Philadelphia and urged the sale of 5,000 batteries. When the President of the company asked about Delco credit Harrison responded that they had no rating at all but that “they had something.”

The upshot was that Harrison had two batteries made to meet Delco needs. One was a twelve-cell weighing sixty-five pounds. When he showed it to Kettering he was told that it was too big. Harrison went back to Philadelphia and asked for lighter ones. By this time his employers were fed up with his enthusiasm for the self-starter and decided to wash their hands of the whole business. Then Harrison played his trump card. He told the head of his concern that if he did not fill the Delco order he would go over to its strongest competitor. This threat carried the day. With his own hands Harrison perfected a light battery that met Delco needs. Before long Harrison was devoting part of his time to Delco interests. Eventually he became a valued part of the organization.

The battery problem was now solved but another difficulty arose. The new and lighter batteries had to be installed and this required time. Meanwhile time was slipping. Leland telegraphed that he was going to Bermuda on a vacation. If the electric self-starter was to go on the 1912 Cadillac, he must test it himself before he sailed. Only three more days remained before the deadline.



To C. F. Kettering the automobile represented a constant challenge and a lifetime field of endeavor.



To prove the sturdy qualities of the self-starter, races were run in which the starter itself was the only motive power allowed.

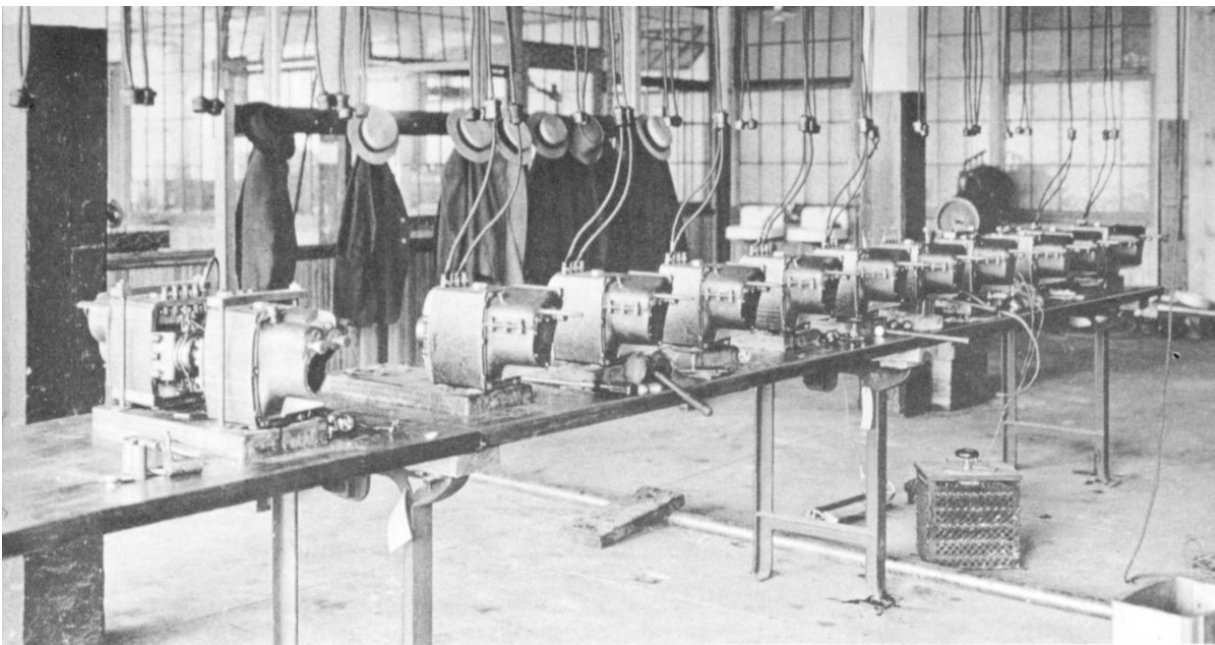
When the completed self-starter seemed ready to be shipped to Detroit something went wrong. For two days and nights Kettering and the “Barn Gang” labored to find the source of the trouble. If the starter did not reach Detroit in forty-eight hours, all was lost. Still the trouble persisted. Finally, late in the afternoon, Kettering sent an S.O.S. to Chryst who was still at the NCR. When he arrived at the barn all he could see of human life was legs protruding from under Kettering’s car which was now the guinea pig. The gang was completely exhausted after the long ordeal of heat, toil, and trouble. “You fellows rest awhile,” said Chryst, “I will try to find out what is wrong.” Wearily Kettering and Anderson hauled themselves out from under the car and before Chryst could say another word fell fast asleep.

Chryst took out the starter mechanism and found that two wires were wrongly connected. In their utter exhaustion Kettering and Anderson had not discovered it. Chryst fixed the wires in short order. The self-starter, now complete, went to Leland by express that night. Once on the 1912 Cadillac, every expectation was realized.

The experimental shop in the barn—the setting of so many headaches—now gave way to a factory. Quarters were rented on the second floor of the Beaver Power Building at the corner of Fourth and St. Clair Streets. Delco

was now a full-fledged and going concern. The advent of the self-starter however required money and machinery. Deeds and Kettering had little of either. With implicit faith in the enterprise, they mortgaged their homes and borrowed money on their life insurance to raise the wherewithal for equipment. In the same year that the self-starter went on the Cadillac, Delco engaged the entire new four-story Beaver Power Building then nearing completion. When Deeds signed the lease, Fred Beaver, owner of the 15 building, said to him:

“You boys will never fill this building making self-starters.” To this Deeds replied: “Just watch us grow.” And grow they did. The second Delco order to O. Lee Harrison was for 15,000 storage batteries, the largest placed up to that time. An enterprise that began in a barn with scant assets save the faith, energy, and inventive genius of the founders, had created an industry whose name carried weight wherever automobiles were made.



Some of the first self-starters manufactured by the newly formed Delco.

Ultimately Delco became part of the vast and far-flung General Motors organization which progressively brought new industries and with them expanded employment to Dayton. The small acorn, planted in the barn, had sprouted to giant proportions.

The business arrangement with Leland developed into a strong and lasting friendship. Father and sons could have had no deeper regard for each other than the dean of motordom and the two young men who had made Delco possible. So satisfactory was the relationship between them that the contract for the ignition system was their first and last. Subsequent dealings involving many millions of dollars were arranged by telephone, telegraph, or orally.

This relationship had an interesting and, so far as the automobile industry is concerned, an historic sequel. Although the episode did not transpire in the barn it was a direct result of all that happened there. As such it is well worth relating.

As Delco grew in prestige and prosperity Deeds and Kettering purchased between them a tract of beautiful rolling country just outside Dayton. 16 Each built a residence on his property. Deeds called his place Moraine Farm because all the hills around Dayton are terminal moraines. One of the first features he installed was a model dairy where the milking was done by electricity. Part of the dairy became the setting of an experiment that re-enforced motor car equipment. The Delco electric ignition, lighting, and starting system had been born in a barn. The Cadillac V-8 was perfected in a cow shed.

The first Delco system had been put on a four-cylinder car. In 1914 some of the large automobile manufacturers advertised that they would produce a six-cylinder car. Leland at once announced that Cadillac would never make a six-cylinder car and gave it wide publicity.

Deeds learned that Leland's plan to meet the six-cylinder car competition was to put a two-speed rear axle on his four-cylinder car. Deeds, as you will recall, had installed a two-speed rear axle on his "Suburban Sixty" and found it wanting. He realized that with such an arrangement Leland would trail in the parade of high-powered cars. Deeds and Kettering viewed the situation with concern so they decided to do something about it. If Leland's competitors were to have a six-cylinder car, why should he not have an eight-cylinder car? At that time an 8-cylinder car was undreamed of so far as American production was concerned. The idea was revolutionary, but

Deeds and Kettering had always done the revolutionary. They now set about doing it again.

The first V-8 car seen in this country was a French De Dion which was exhibited at the New York Automobile Show. Deeds and Kettering purchased it and had it shipped out to Moraine Farm. They found it clumsy and unsuitable for study or adaptation. If Leland had ever seen it he would never have fallen in with the suggestion for a Cadillac V-8.

Deeds cast about for a suitable experimental engine. He knew that E. J. Hall of the Hall-Scott Motor Company in California—the same Hall who was destined to have such an important part in the designing of the Liberty engine—had built an eight-cylinder aeroplane engine and installed it in a chassis. Perhaps this might serve the purpose. Deeds wired Hall to send it by express. “Do you mean express?” wired Hall in reply. “Yes,” answered Deeds.

The engine and chassis were shipped to Cincinnati where Kettering and Chryst, who was now a part of the Delco organization, picked it up. They drove it to Dayton, arriving in the early hours of the morning wet, cold, and mud-splattered. The engine was put on a Stevens-Duryea car owned by Kettering. It was christened “Old Steve” and became a guinea pig to rank in motor car history with “Suburban Sixty.”



Almost 40 years after the invention of the self-starter, Kettering and Deeds sit across from each other at the old desk which they formerly shared. The occasion was "Deeds Day" at Denison University, where this desk is now preserved.

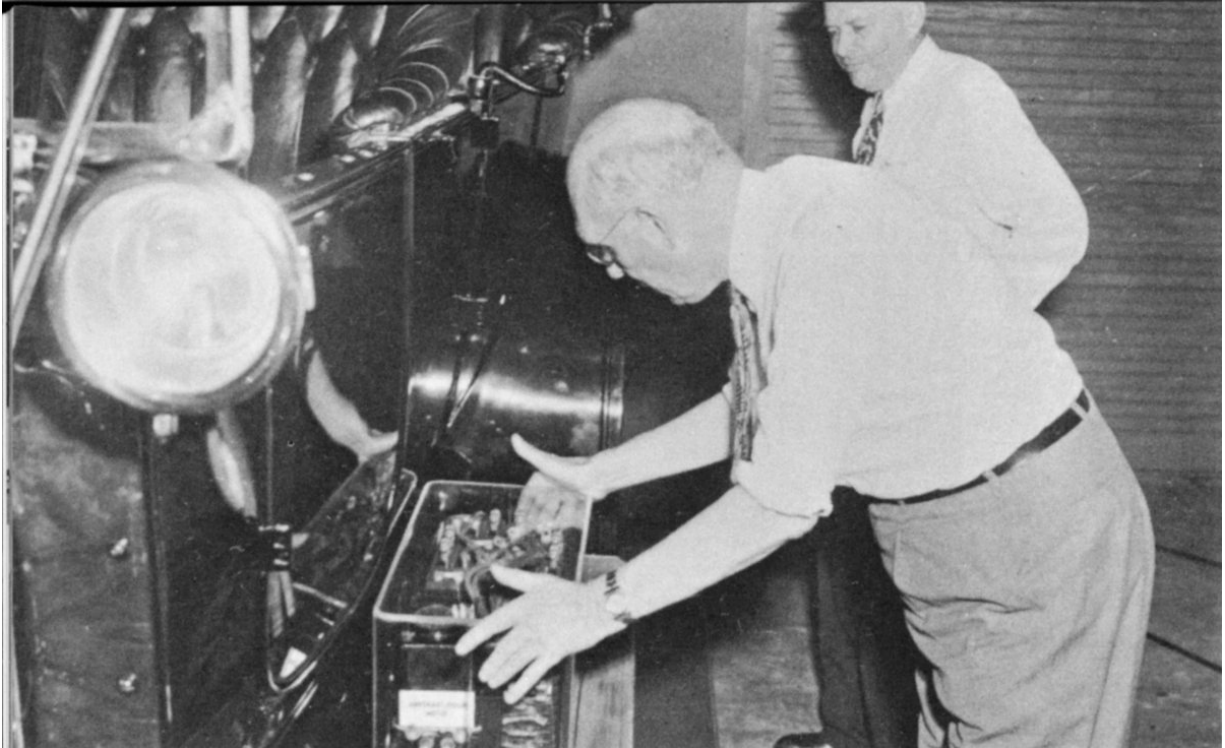
Detroit is a rumor factory. If the work had been carried on there, the whole city would have known about it. It was therefore necessary to carry out the experiments in Dayton and in secrecy. Deeds turned the cows out of his barn and used it as a machine shop. All the machinery was installed at night and most of the work was done while Dayton slept. The secret was well kept. Deeds and Kettering improved the Hall engine and then took "Old Steve" to Toledo, where Leland gave it a drastic tryout. It stood up under every test, proving that the V-8 principle was right.

Leland now entered enthusiastically into preparations for his Cadillac V-8. Secrecy was still necessary. He therefore rented a shop in Worcester, Massachusetts, and brought over D. McCall White who had built the British Napier Eight to design the car. White did his work well. In 1914 the Cadillac V-8 was launched with proper eclat, enhancing the prestige of American-made automobiles.



Thousands of self-starters roll down the production line in modern factory.

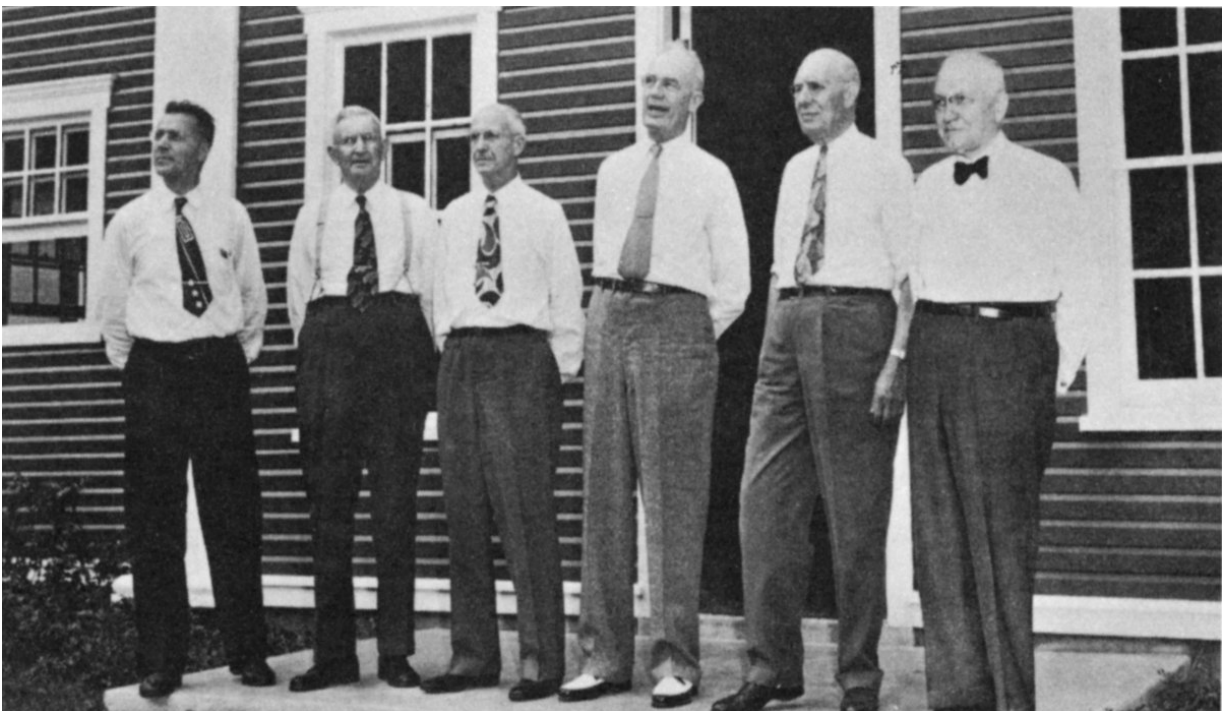
18



C. F. Kettering and C. E. Wilson, then president of General Motors, inspect the battery on the 1912 Cadillac during a 1949 visit to the barn.

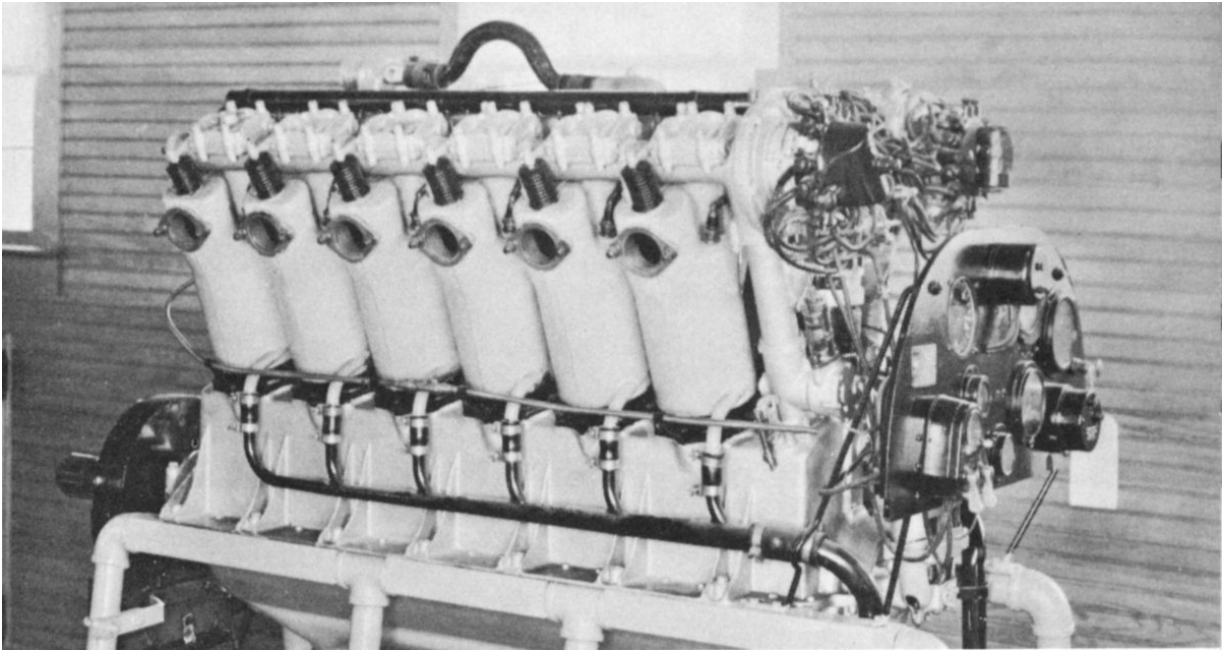
If Deeds and Kettering had not carried out their experiments in the cow barn at Moraine Farm, the Cadillac V-8 might never have come into existence. Certainly it would have been long delayed. It was a labor of love on the part of Deeds and Kettering in gratitude for what Leland had done for them when Delco was little more than an inspiration. They never accepted a cent of compensation.

The Delco achievement stands as a thing apart in the development of American productive enterprise. It created a new industry, revolutionized the use of the automobile, and launched Deeds and Kettering on the road to wider and equally constructive fields. One became an outstanding industrial leader and builder; the other rose to be the greatest industrial scientist of his time.



The 40th anniversary of the invention of the starter found Kettering, Deeds and other members of the old “Barn Gang” previewing the displays in Deeds Barn.

The Liberty Engine



The Liberty engine, a milestone in the development of American aviation, is among the historic exhibits in Deeds Barn.

The Liberty Engine, one of the outstanding achievements of World War I, represents an engineering and manufacturing triumph that contributed much to man's mastery of the air.

Conceived in less than six weeks during a time of crisis, it was the finest aircraft power plant of its time, and for a decade thereafter remained the nucleus around which aerial advances were made.

The story of the Liberty engine begins in 1917, shortly after the United States entered World War I. Air power, even then, had become a vital factor in military operations, and the hard-pressed Allies desperately needed a quantity-produced aircraft engine that would wrest control of the skies from the Central Powers. They turned to the United States and its industrial might for a solution to this critical problem.

How productive this appeal was to be was borne out in the events that followed. At the direction of Colonel Deeds, who was then serving as Chief of Aircraft Procurement, two automotive engineers literally locked themselves in a Washington hotel room and went to work. The date was May 29, 1917.

Colonel Deeds' instructions had been brief but to the point. He cautioned the engineers, J. G. Vincent and E. J. Hall, that the engine they designed had to be light in weight in proportion to power, and adaptable to mass production. Then he added a final admonition: the engine must embody no device that had not already been tested and proved in existing engines. There was no time for experimentation.

Within forty-eight hours Vincent and Hall had come up with the rough design of the engine. Three days later they had incorporated in the design all of its major features, and shortly thereafter the nation's auto manufacturers had started production of parts for five 8-cylinder engines and five 12-cylinder engines. 20

Deeds had been shooting for completion of the first trial engine by July 4. This goal, which seemed to many the strongest form of wishful thinking, was more than met in what will remain as one of the classic achievements in American production annals. Impossible as it had seemed, a complete 8-cylinder engine was delivered to the nation's Capital on July 3, a day before the deadline and less than six weeks after the design was drawn. The first 12-cylinder engine passed its tests the following month.

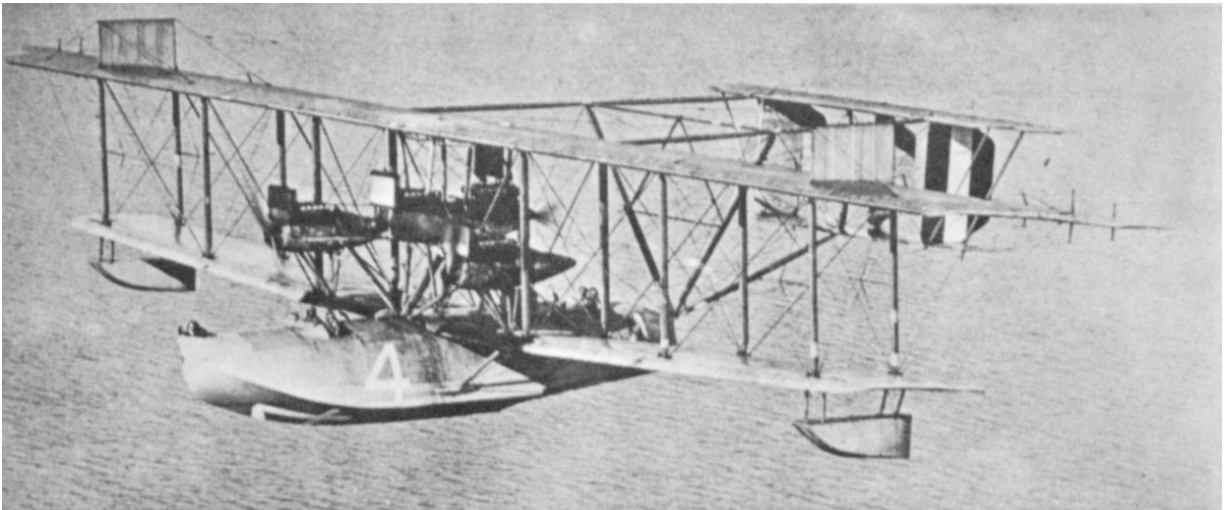
In the short span of a year from the date when its designers first met, 1,100 Liberty engines had been produced and put into service. By the time the Armistice was signed, 24,475 engines had been turned out and daily production of the V-12 had climbed to the staggering figure of 150 every 24 hours. The magnitude of this engineering, manufacturing and management feat is pointed up by the fact that a host of small jigs, tools and fixtures were required to build every part of the complicated engine.

Newton D. Baker, Secretary of War, said of the project: "I regard the invention and rapid development of this engine as one of the really big accomplishments of the United States since its entry into the war.... The

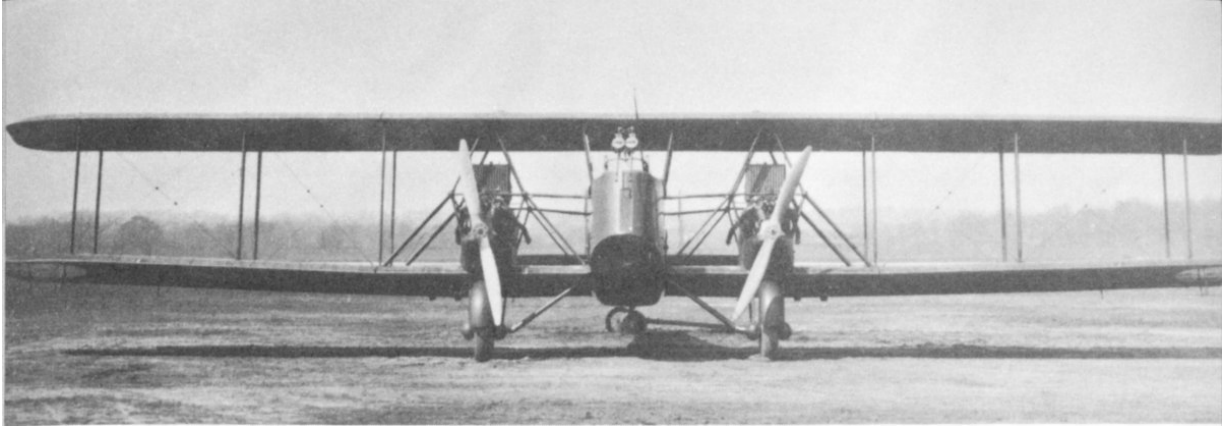
story of the production of this engine is a remarkable one. Probably the war has produced no greater achievement.”

Although the Liberty engine became a potent factor in World War I, its greatest achievements were to come in the years following, when for the first time aviation’s horizon was lifted above the level of county-fair stunt exhibitions and the foundation laid for commercial aviation.

It was the Liberty engine that first enabled man to conquer the ocean in an airplane. Four of the 440-horsepower engines powered the famous U. S. Navy NC-4, which in 1919 became the first plane to fly the Atlantic.



Aerial mastery of the Atlantic was first achieved in 1919 in the Liberty-powered NC-4, shown in this U. S. Navy photograph.



The Martin MB-2 was used by General Mitchell in the 1921 tests which indicated the destructive ability of big bombing planes. Liberty engines supplied the power.

Two years later, when General “Billy” Mitchell conducted his historic and dramatic demonstration of the potency of air power, Liberty-powered planes were called upon to do the job. The big bombers, as Mitchell had predicted, sank a 22,000-ton German battleship in only twenty-five minutes.

In April, 1923, the Liberty was again in the headlines. Two Army pilots, Lieutenants Oakley Kelly and John Macready, flying a Fokker airplane powered with a Liberty engine, stayed aloft for thirty-six consecutive hours, shattering all previous endurance marks. Later the same year they completed the first non-stop transcontinental flight, again using the Liberty-powered Fokker.

The following year still another achievement was added to the Liberty’s long series of successes. Two Douglas Cruisers, powered with Liberty engines and prepared for the flight at Dayton’s old McCook Field, became the first planes to circle the globe, covering 26,345 miles in 363 hours of actual flying time.



The famous De Havilland 4, British-designed observation plane of World War I. The ship shown here is an American version, built by the Dayton Wright Airplane Company in 1918 and powered with a Liberty engine. At left is Orville Wright, consultant for the company, shown with Howard Rinehart, a test pilot.

22



The first planes to complete a trip around the world were two Douglas Cruisers, one of which is shown above. There again, the Liberty engine helped establish a memorable record in aviation history.

Besides holding at one time the world's speed and altitude records, the Liberty also was the engine that first made the concept of air-mail service more than just a dream. Charles A. Lindbergh was among the pioneer pilots of the early 1920's who flew Liberty-powered De Haviland mail planes along relatively uncharted air lanes through all types of weather.

Thus, the war-born Liberty became the cornerstone of American commercial aviation and the forerunner of today's vast and complex network of aerial transit. Years ahead of its time, the Liberty engine stands today as an outstanding example of American productive genius. Born in time of war, it played an important role in time of peace. In aviation's hall of fame, the Liberty engine is assured a place of honor for decades to come.



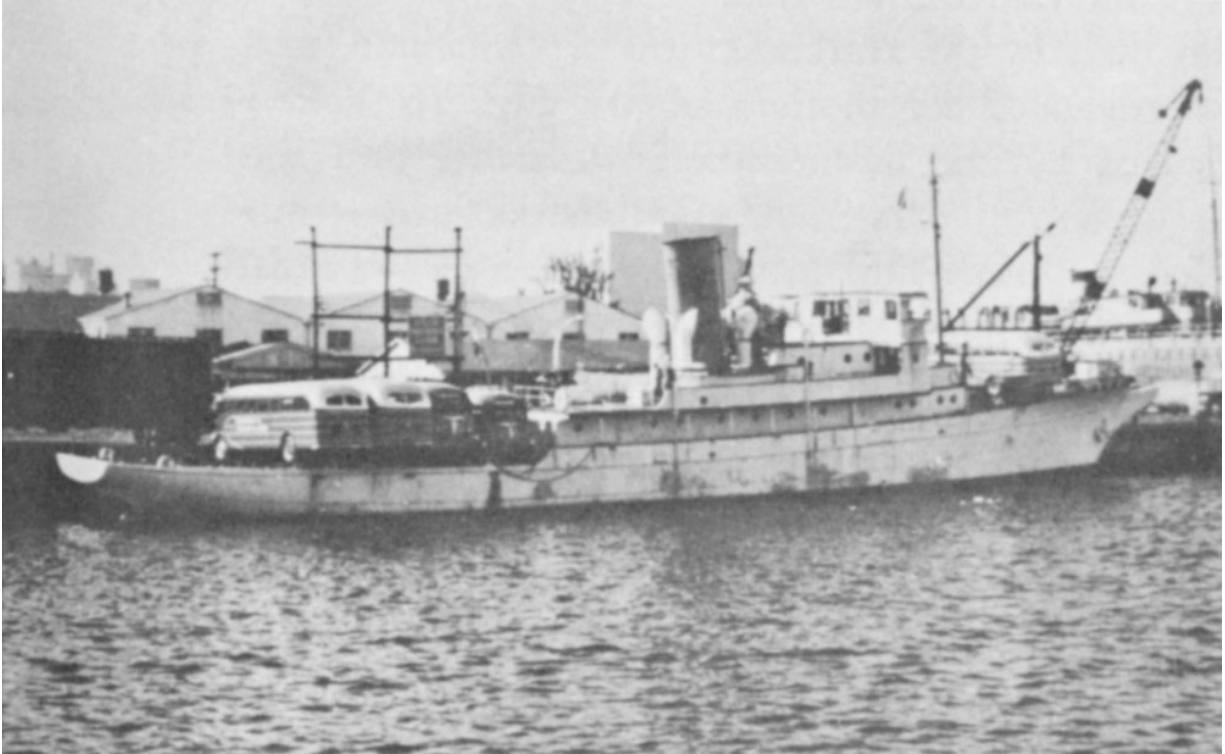
Lieutenants Kelly and Macready, dwarfed by the giant Fokker T-2 in which they established a new endurance record in 1923, are shown here with the fuel required for the famous flight.

The “Lotosland”



This is a model of the “Lotosland,” the yacht formerly owned by Colonel Deeds. It was placed in the exhibit as an example of maritime transportation.

The “Lotosland” was for a number of years the second home of Colonel and Mrs. Deeds. It was a trim vessel of 206 feet, built to Colonel Deeds’ specifications. While it was never used for long ocean voyages, it was fully capable of making them. The “Lotosland” included many innovations, one of the most important of which was the amphibian airplane which it carried. She was the first ship, other than a naval vessel, to be so equipped. When the United States entered World War II, the Navy requisitioned the “Lotosland” and converted her into a combat ship for anti-submarine patrol. She made a distinguished war record and accounted for at least one submarine. After the war the “Lotosland” was disposed of to private shipping interests and, at last reports, was plying between Miami, Florida, and South America.



This picture of the “Lotosland” was taken by Porter B. Chase, who knew her in prewar days and recognized her in port at Miami on January 10, 1950. She is loaded with five busses, destined for delivery to South America.

The “Cincinnati”



Replica in miniature of the “Cincinnati,” first locomotive to pull a passenger train into Dayton on the old Cincinnati, Hamilton & Dayton Railroad, which later became part of the Baltimore & Ohio system.

The “Cincinnati,” prototype of the replica in miniature on display in Deeds Barn, was typical of the locomotives that in the 1850’s were beginning to establish a new transportation network across Ohio and the Middle West.

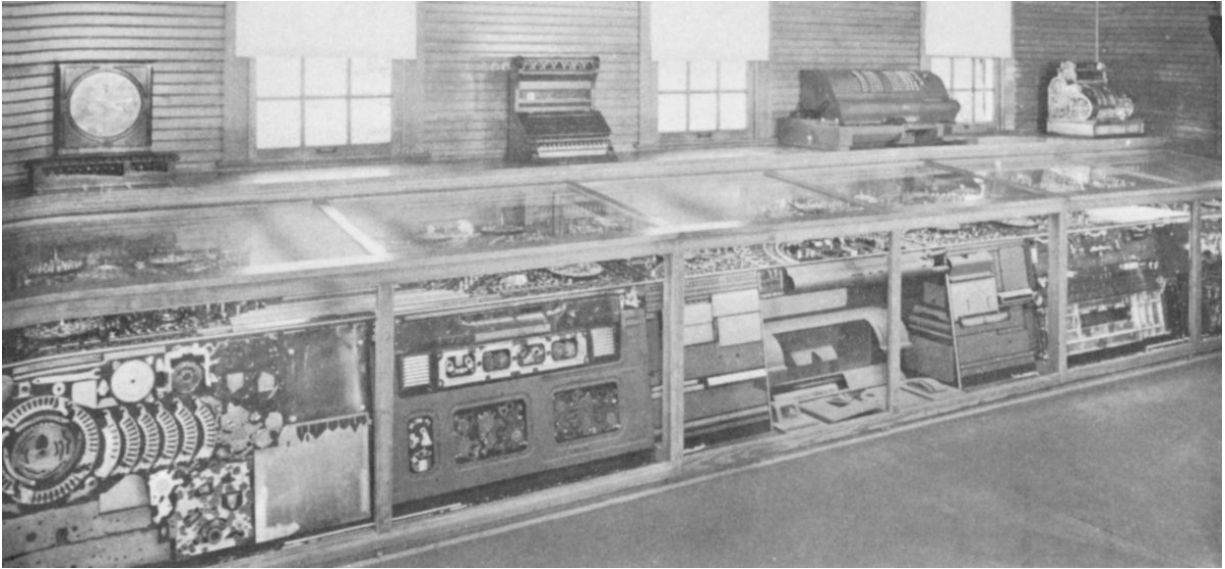
These locomotives and their successors were to doom the canal system which played such an important role in development of the West. Although the Miami and Erie Canal—which linked Dayton to Cincinnati and Toledo—had its peak year in 1851 when the “Cincinnati” first steamed into Dayton, canal operating expenses outran revenue only five years later. In

1877, official operation of the canal ended. In the march of progress, the railroad had won.

The “Cincinnati” was built by the Harkness firm of Cincinnati, and was one of the most advanced locomotives of its day. Its honest functionalism, as evidenced by the businesslike cow-catcher and the monstrous stack designed to trap dangerous wood sparks, makes this type of locomotive, called an “American,” a favorite with railroad fans.

How long the “Cincinnati” remained in service with the C.H.H.&H.D. Railroad, and its ultimate fate are not known, but the colorful locomotive and others of its kind made possible significant gains in man’s never-ending quest for better transportation.

The Cash Register Display



Display of the 20,000 parts which make up a modern National Accounting Machine, outgrowth of the early cash register.

Inasmuch as C. F. Kettering and E. A. Deeds began their careers with The National Cash Register Company, it seemed fitting that some representation of that company's products should be included in the barn display. The display includes early models of National cash registers and an impressive exhibit of the 20,000 parts which go to make up a modern National accounting machine.

The cycle of events moves in unpredictable and often unexpected ways. When E. A. Deeds left NCR to help develop the self-starter business, he probably never expected to be a part of that organization again. Yet in 1931 he was named Chairman of the Board of that company and subsequently played an important part in its very considerable development.

When Deeds and Kettering first went with NCR, the line of machines which made up the company's products was limited indeed. It was they who took the first steps toward an electrically operated cash register. The possible

entrance of the company into the accounting machine field had not even been considered. Employment at the local plant was about 2,400 people.

Today The National Cash Register Company derives almost as much of its business from the sale of accounting machines as from the sale of cash registers. Through continuous research and advanced engineering, it has created machines which meet the needs of business wherever money is handled or records are kept.

The Company now employs 13,000 people at the Dayton plant. With complete accuracy, the cash register bell is often referred to as “The Bell Heard 'Round the World.”



Patterson Boulevard, on which Carillon Park is located, is one of Dayton's busiest arteries of traffic. There are two reasons why it is fitting to include this picture of teeming traffic with the story of the self-starter. If the self-starter had not been invented, the automobile would never have reached the place that it has in the lives of the people of this country. Continued dependence on the old hand crank as a means of starting would have been an almost unsurmountable handicap. It is also interesting to know that many of the cars which contribute to the stream of traffic on Patterson Boulevard are those of employees of NCR and of General Motors. Deeds and Kettering started their business careers with NCR and were responsible for bringing the first unit of General Motors to Dayton.

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